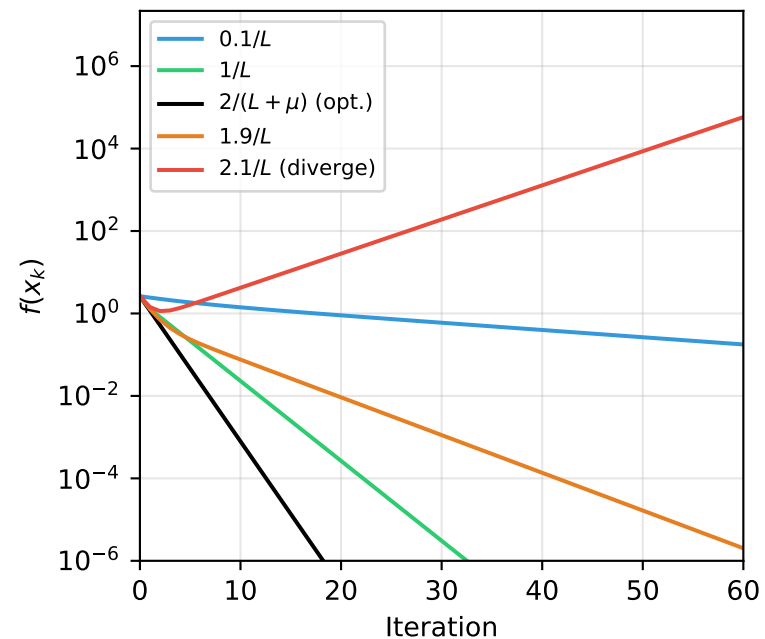
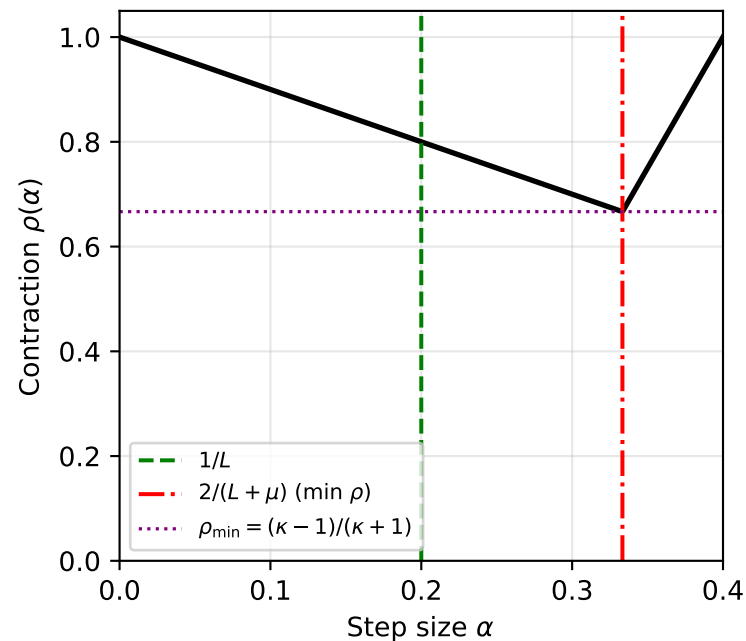


GD/Nesterov/Heavy Ball: Step-Size Theory, Bounds, and Optimality

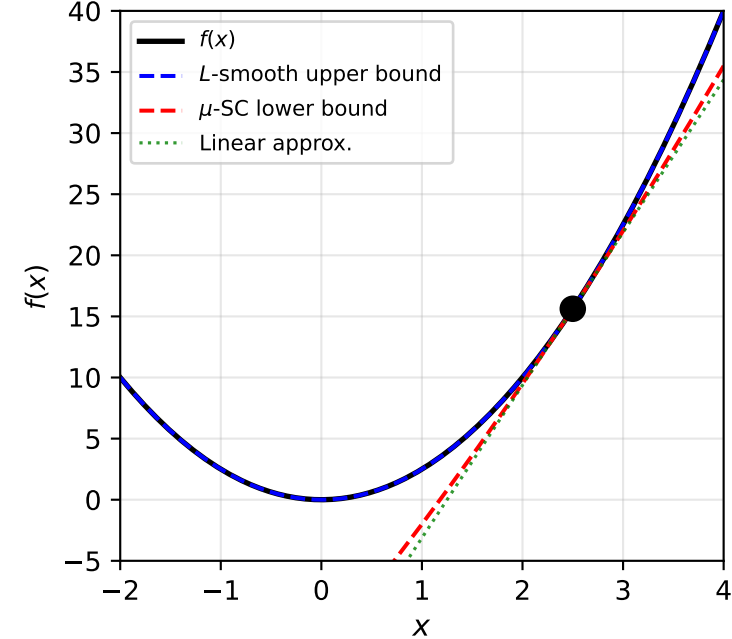
GD Step-Size Regimes ($\kappa = 5$)



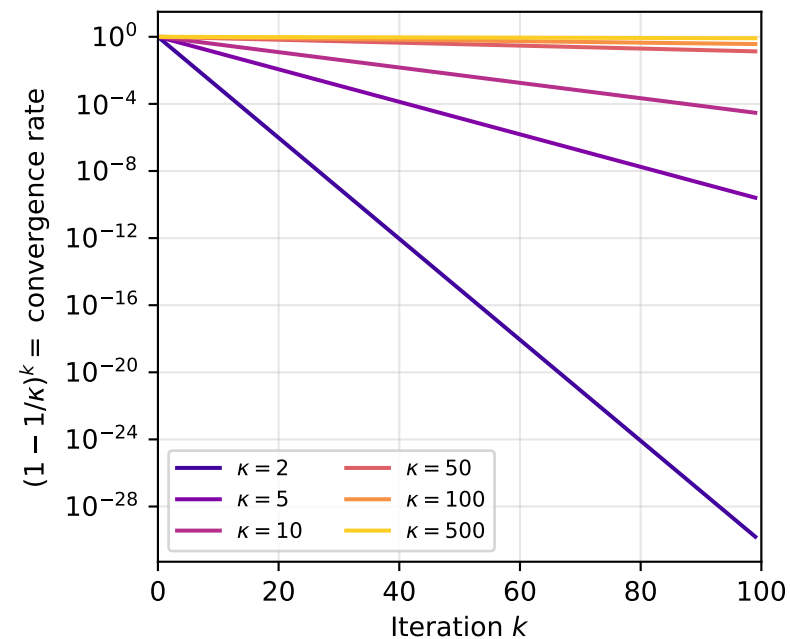
GD Contraction Factor



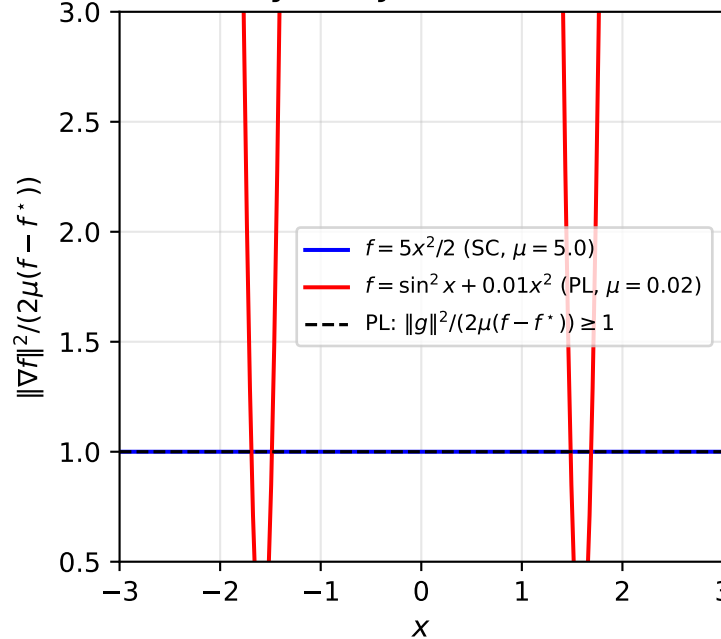
L-smooth and SC Function Bounds



GD Convergence Rate vs κ



Polyak-Łojasiewicz Ratio



Lower Bounds: Optimality of Nesterov/HB

